Fleexa

2026

Virtual IoT Fleet Management System

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Motivation

The driving force and necessity behind the project.

The Cost Barrier

Physical IoT labs are expensive to set up and maintain.

The Risk Factor

Testing on real hardware like gas sensors or high-voltage actuators can be risky for beginners.

The Scalability Issue

It is difficult to scale physical hardware for large numbers of students or trainees.

Objectives

The main goals we aim to achieve with Fleexa.

01

Build a fully simulated IoT environment
(No hardware required).

02

Implement a complete Cloud-Native architecture (AWS).

03

Provide real-time telemetry and control via a Mobile App.

04

Achieve zero-cost deployment using free tier services.

05

visualize device data through interactive dashboards and insights.

Business Plan

Market analysis and business model.

The barrier:
60% of IoT Pilots fail
Due to $50K cost

The Solution:
Production-Grade
Simulation Scale at low
cost

Market Opportunity:
$7.8B | 18.9% CAGR

Competitive Dedition

GEOTAB

samsara

Tamra
تمرة

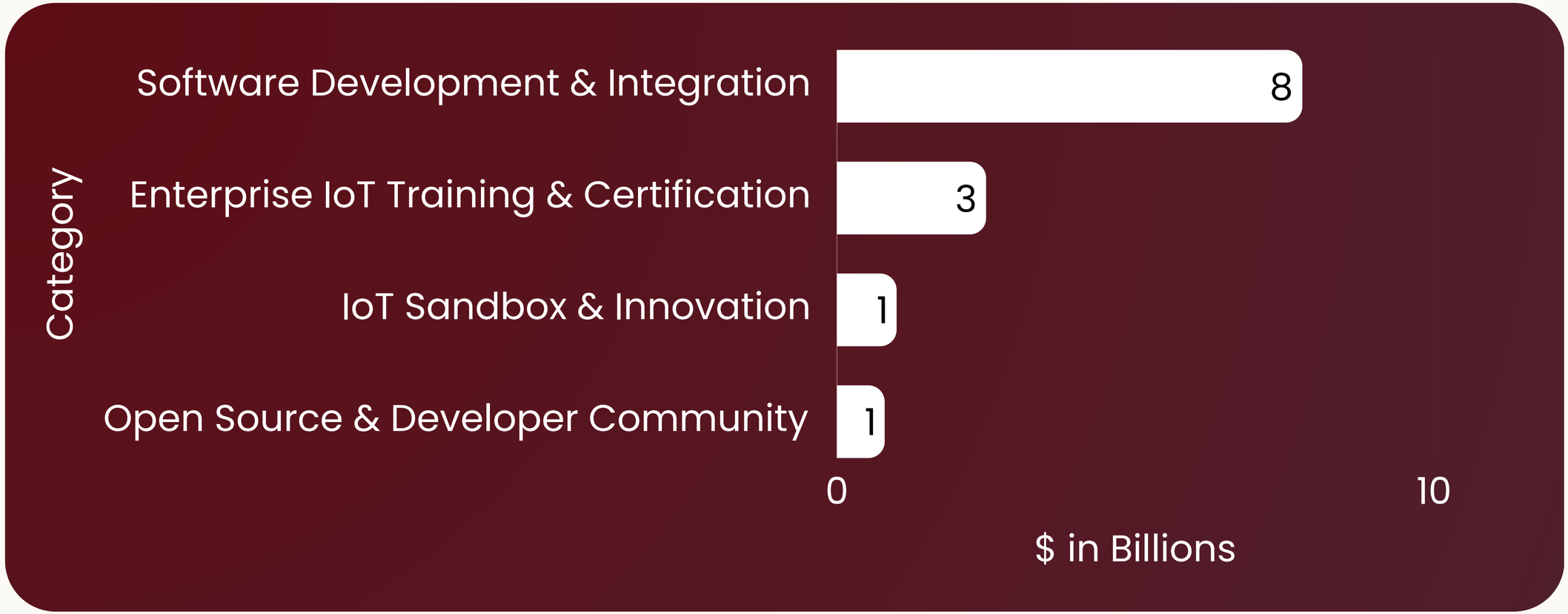
$0
Cost

<1hr
Deploy

7 - Layer
Architecture

Business Plan

Market analysis and business model.



Techonlogies

The tools powering the application.

Flutter

Figma

Python

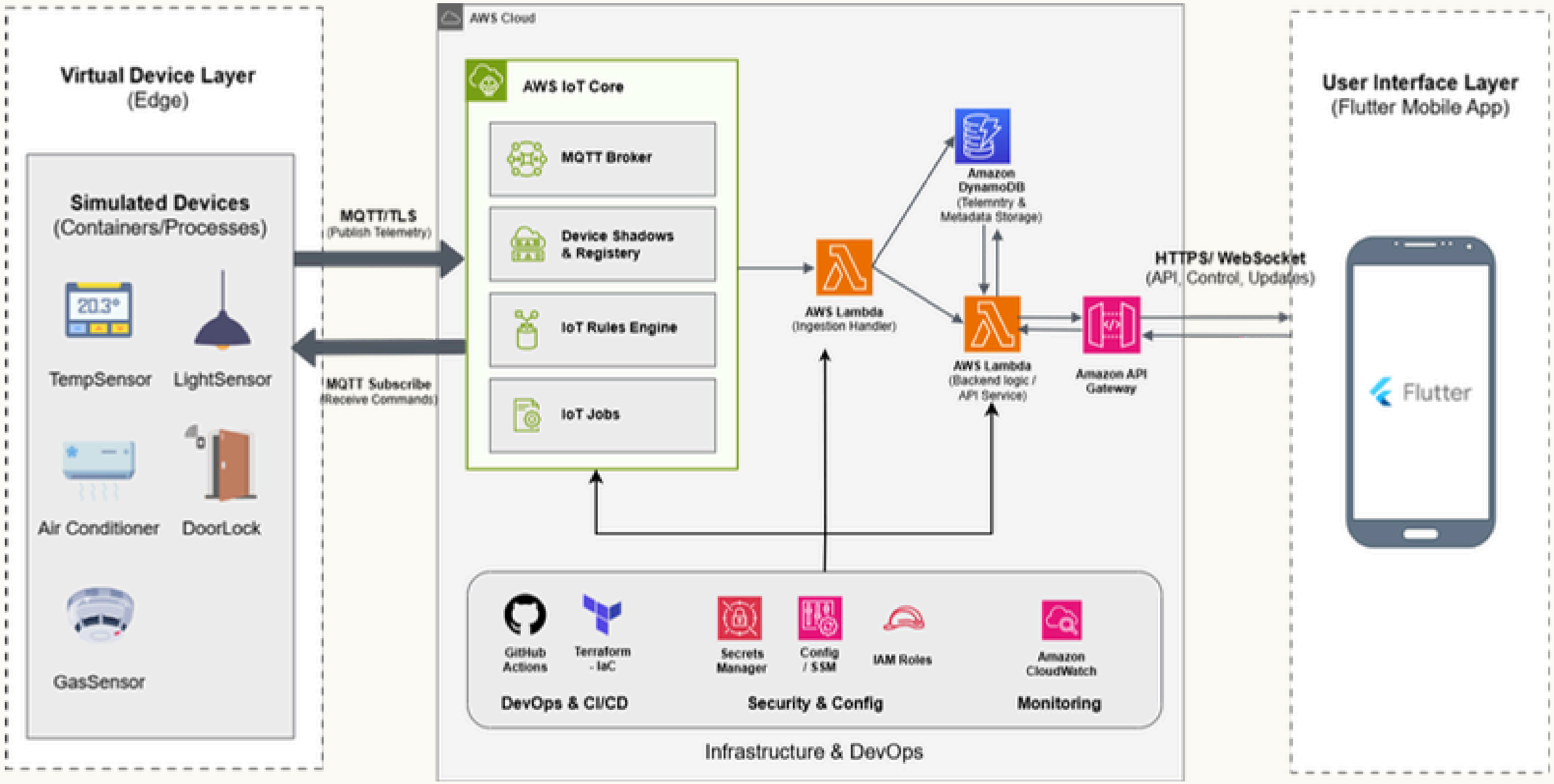
Go

Docker

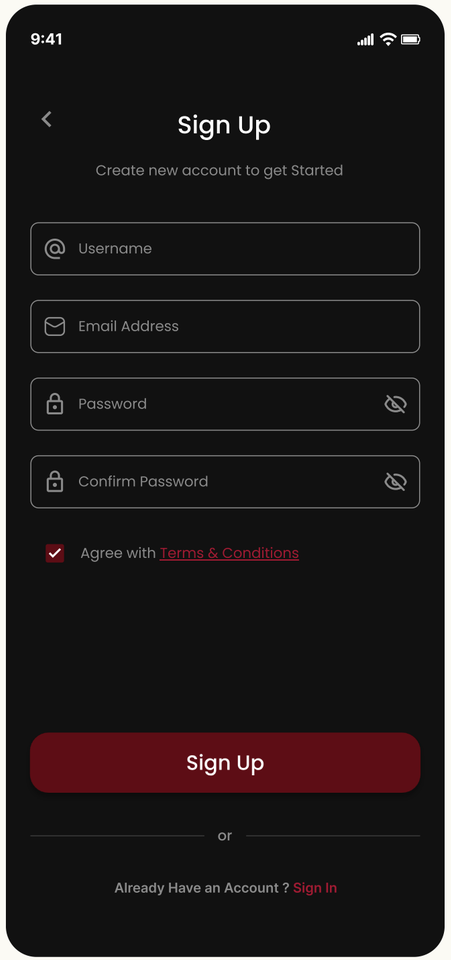
GitHub

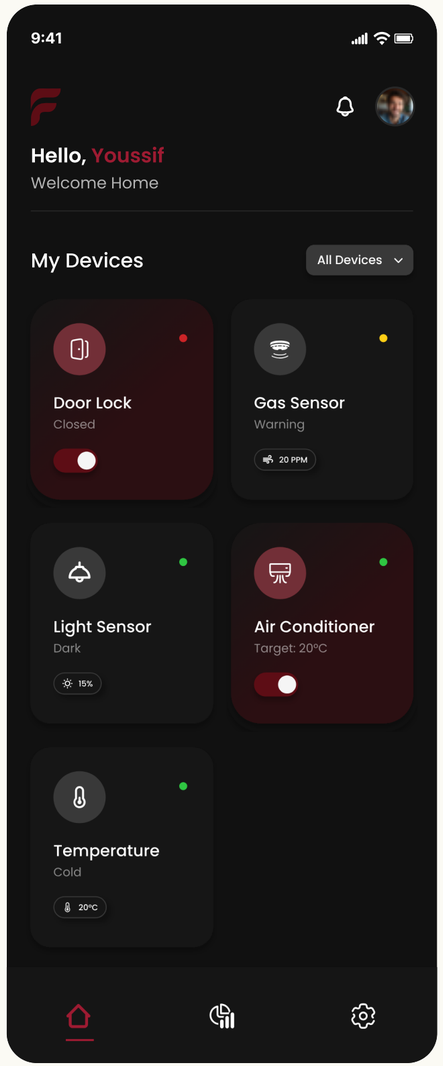
AWS

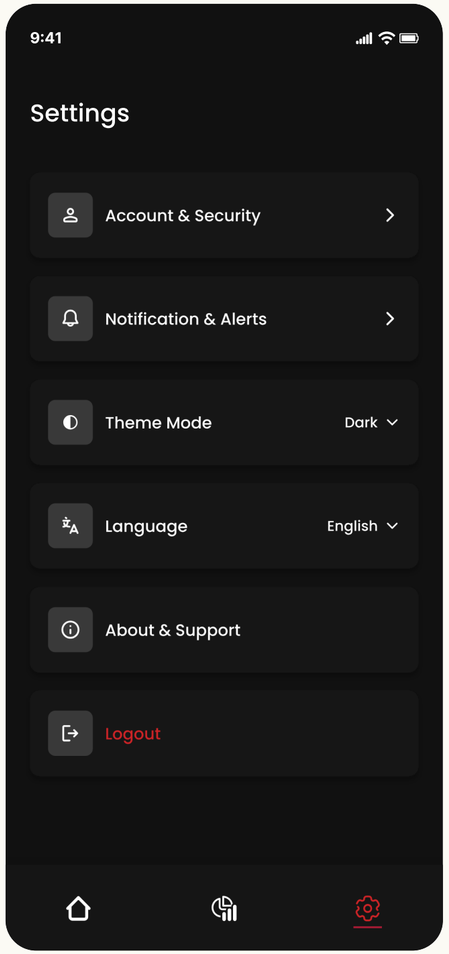
High Level Architecture

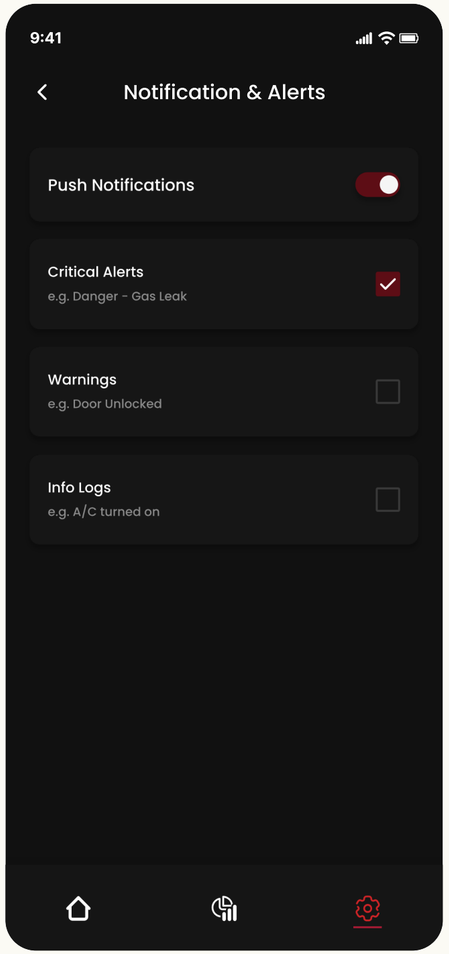


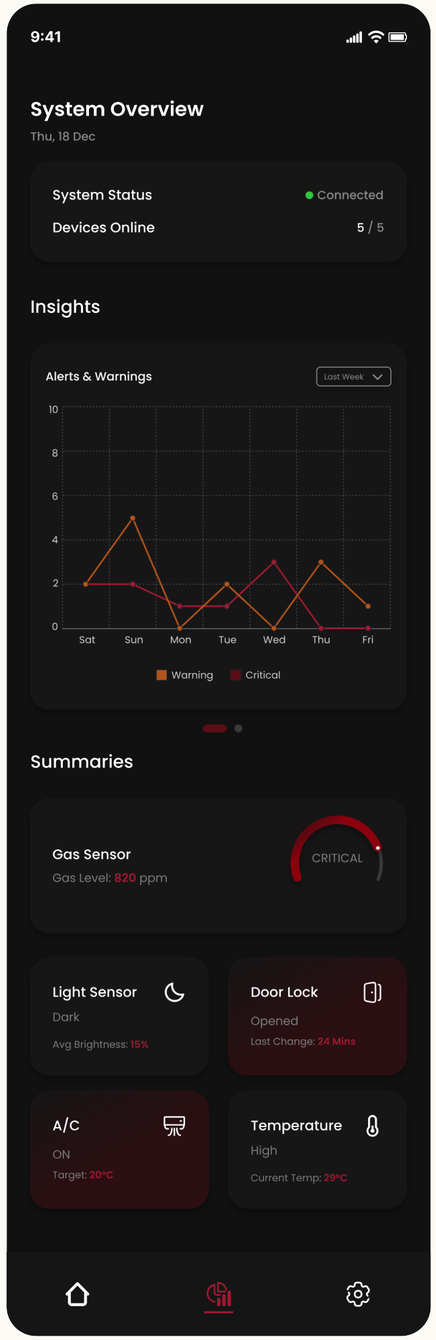
UI/UX


9:41
Sign Up
Create new account to get Started
Username
Email Address
Password
Confirm Password
Agree with Terms & Conditions
Sign Up
or
Already Have an Account ? Sign In

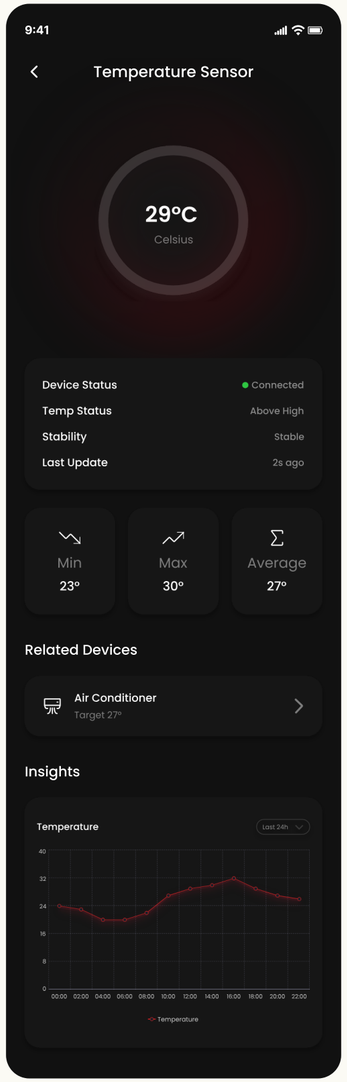

9:41
Hello, Youssif
Welcome Home
My Devices
All Devices
Door Lock
Closed
Gas Sensor
Warning
20 PPM
Light Sensor
Dark
15%
Air Conditioner
Target: 20°C
Temperature
Cold
20°C

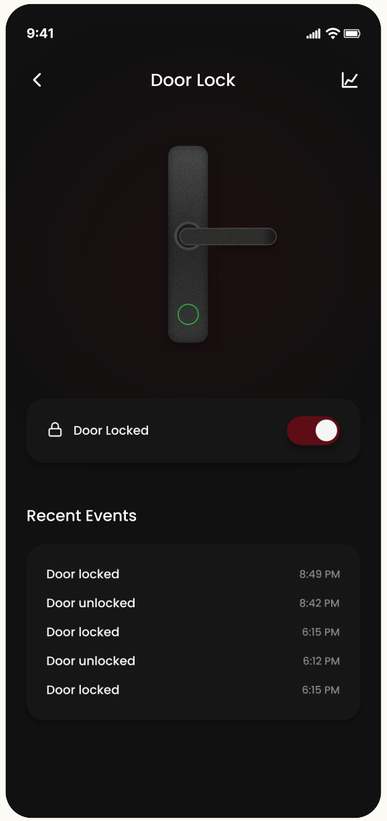

9:41
Settings
Account & Security
Notification & Alerts
Theme Mode
Dark
Language
English
About & Support
Logout

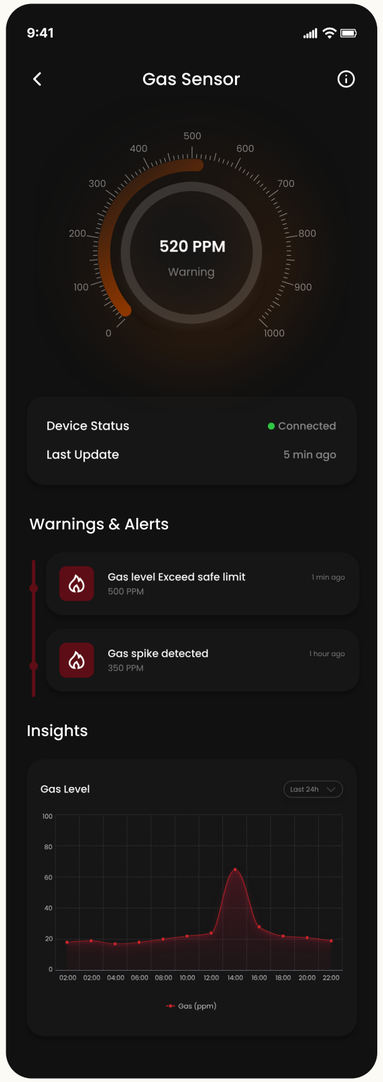

9:41
Notification & Alerts
Push Notifications
Critical Alerts
e.g. Danger - Gas Leak
Warnings
e.g. Door Unlocked
Info Logs
e.g. A/C turned on

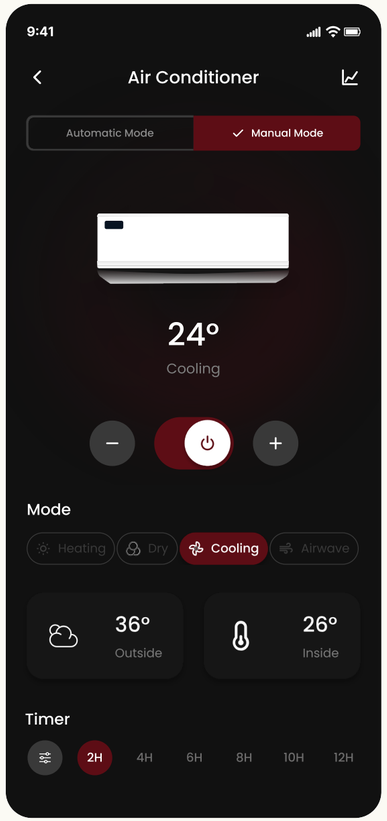

9:41
System Overview
Thu, 18 Dec
System Status
Connected
Devices Online
5 / 5
Insights
Alerts & Warnings
Last Week
10
8
6
4
2
0
Sat
Sun
Mon
Tue
Wed
Thu
Fri
Warning
Critical
Summaries
Gas Sensor
Gas Level: 820 ppm
CRITICAL
Light Sensor
Dark
Avg Brightness: 15%
Door Lock
Opened
Last Change: 24 Mins
A/C
ON
Target: 20°C
Temperature
High
Current Temp: 29°C

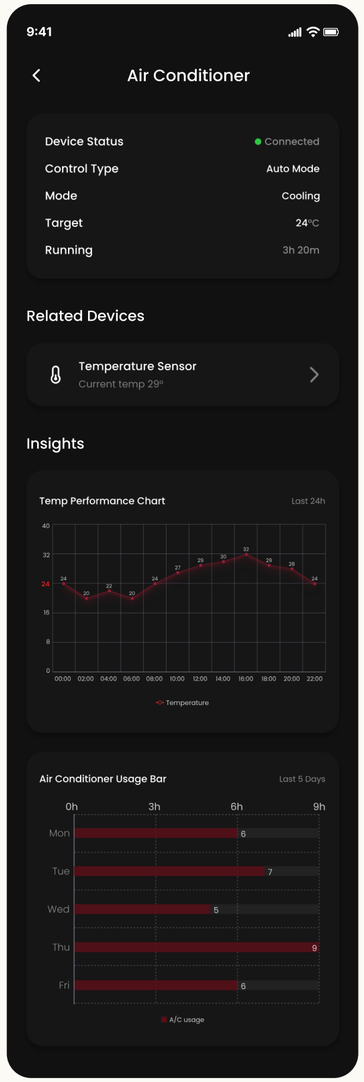
UI/UX

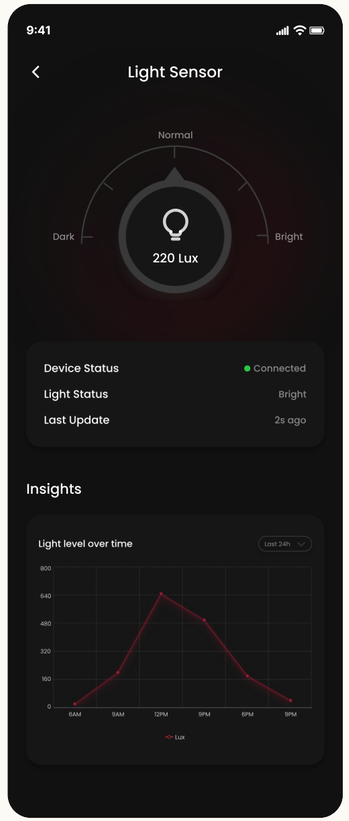

9:41
Temperature Sensor
29°C
Celsius
Device Status
Connected
Temp Status
Above High
Stability
Stable
Last Update
2s ago
Min
23°
Max
30°
Average
27°
Related Devices
Air Conditioner
Target 27°
Insights
Temperature
Last 24h
Temperature


9:41
Door Lock
Door Locked
Recent Events
Door locked 8:49 PM
Door unlocked 8:42 PM
Door locked 6:15 PM
Door unlocked 6:12 PM
Door locked 6:15 PM


9:41
Gas Sensor
520 PPM
Warning
Device Status
Connected
Last Update
5 min ago
Warnings & Alerts
Gas level Exceed safe limit
500 PPM
1 min ago
Gas spike detected
350 PPM
1 hour ago
Insights
Gas Level
Last 24h
Gas (ppm)

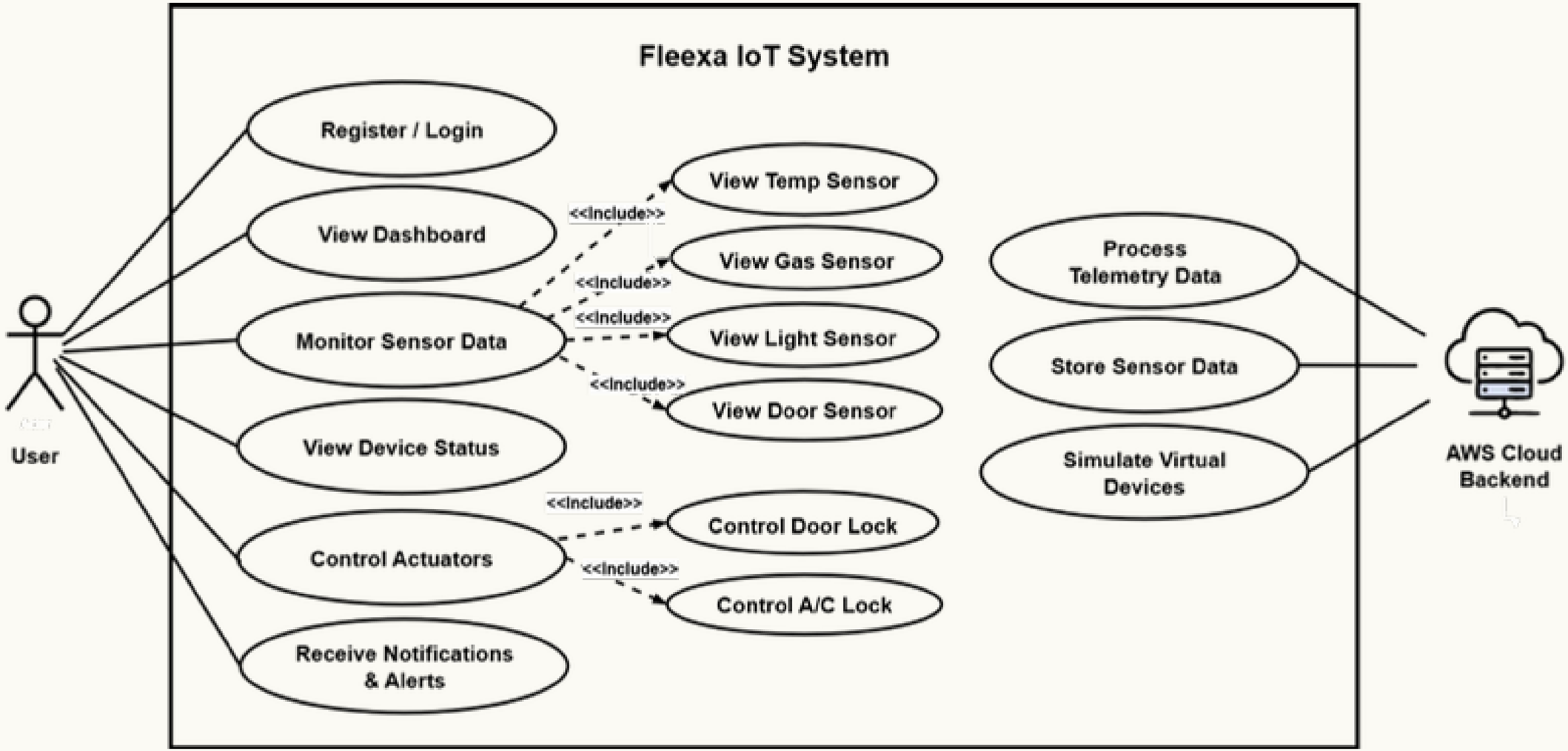

9:41
Air Conditioner
Automatic Mode
Manual Mode
24°
Cooling
Mode
Heating
Dry
Cooling
Airwave
36°
Outside
26°
Inside
Timer
2H
4H
6H
8H
10H
12H


9:41
Air Conditioner
Device Status
Connected
Control Type
Auto Mode
Mode
Cooling
Target
24°C
Running
3h 20m
Related Devices
Temperature Sensor
Current temp 29°
Insights
Temp Performance Chart
Last 24h
Temperature
Air Conditioner Usage Bar
Last 5 Days
0h
3h
6h
9h
Mon
Tue
Wed
Thu
Fri
A/C usage


9:41
Light Sensor
Normal
Dark
Bright
220 Lux
Device Status
Connected
Light Status
Bright
Last Update
2s ago
Insights
Light level over time
Last 24h
Lux

Use Case Diagram

Visualizing how users interact with the system.



Feature: Remote Control & Actuation

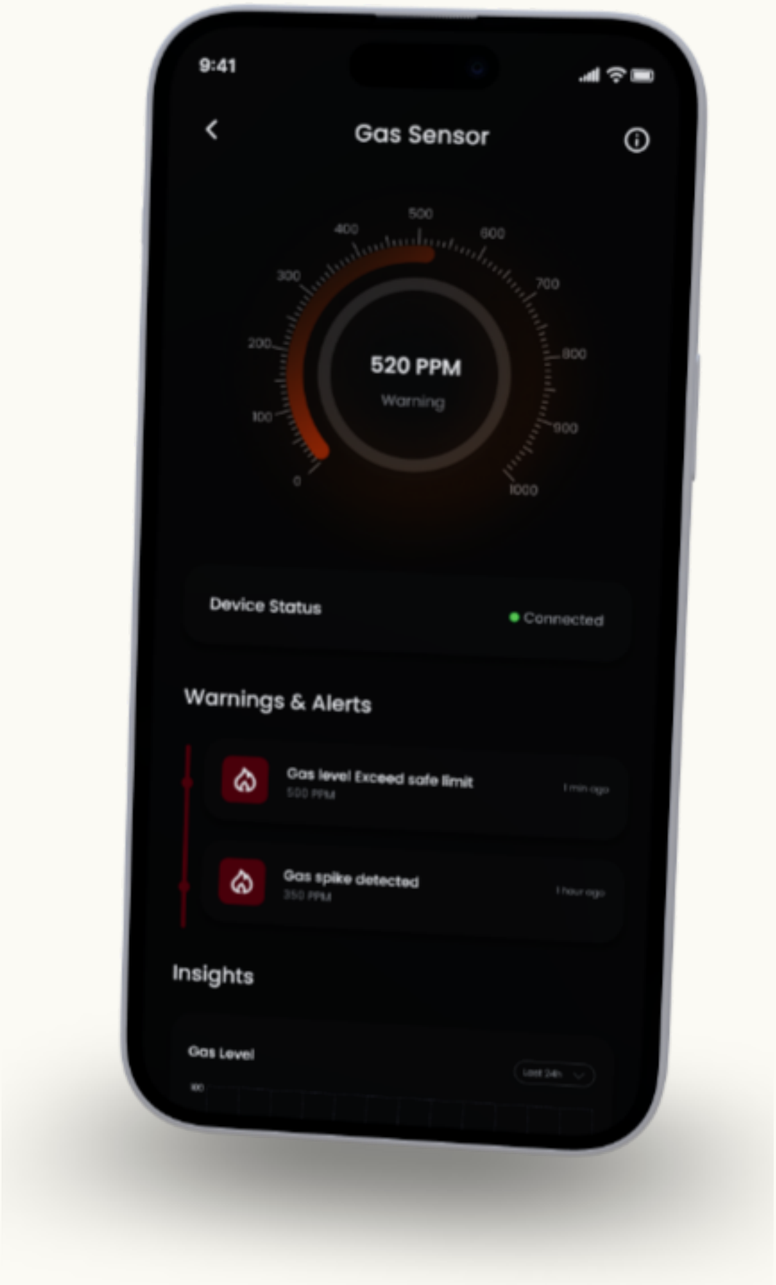
The user sends commands from the mobile app to control virtual devices, such as unlocking the Door or turning on the A/C. The command moves through the cloud to the device, executing the action instantly just like a real smart home.

Feature: Real-time Monitoring

The system simulates virtual sensors (Temperature, Light, Gas) that continuously send telemetry to AWS. The user monitors these readings live on the mobile dashboard to ensure safe environmental conditions.

Feature: Critical Safety Alerts

If for example, the Gas sensor detects a leak or the Door sensor reports an unauthorized breach, the system processes this data in the cloud and immediately sends a notification to the user’s phone.



Feature: Historical Data Analytics

The user selects a specific device (e.g. the Light Sensor) and a time range. The app retrieves stored historical data stored in the AWS database and displays it as charts, providing a clear vision for users to visualize and monitor device performance over time.

Future Plan

The roadmap for further development & expansion.

- **Mobile App Development**
  Convert our designs into Flutter application.
- **Connect the Backend**
  Link the application to our AWS infrastructure.
- **Monitor System Health**
  Establish automated system monitoring and alerts.
- **Launch Data Streams**
  Enable our virtual devices to produce real-time readings.

THANK YOU